AI-504: Deep Learning
Sample Questions

Q1. In a cricket match, a player's performance is represented as:

$$x = [\text{runs}, \text{wickets}, \text{catches}]$$

(a) What type of mathematical object is this?
(b) If $x \in R^3$, what does 3 represent?

Q2. Explain why linear activation functions are insufficient when classifying non-linearly separable football match data.

Q3. Write the gradient descent weight update rule. What happens if:

- Learning rate is very small?
- Learning rate is very large?

Q4. Write the general form of a simple linear regression model.

Q5. State any three properties that a norm must satisfy.

Q7. If net input for a cricket match predictor is:

$$net = 1.2$$

Compute:

$$\sigma(net) = \frac{1}{1 + e^{-net}}$$

Q8. Given:

$$W = 4, \quad \frac{\partial J}{\partial W} = 1.2, \quad \eta = 0.1$$

Compute the updated weight.

Q9. A football match predictor gives the following results:

	Predicted Win	Predicted Lose
Actual Win	40	10
Actual Lose	8	42

Compute:

- Total number of matches
- Accuracy

Q12. During training of a cricket score predictor:

$$\frac{\partial E}{\partial out} = 0.6$$
$$\frac{\partial out}{\partial net} = 0.25$$
$$\frac{\partial net}{\partial w} = 0.5$$

Using chain rule, compute:

$$\frac{\partial E}{\partial w}$$

Q13. A regression model is given by:

$$y = 4.162 + 15.509x$$

where x = number of practice sessions.

Predict y when $x = 3$.

Q14. Given:

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

- Compute $\sigma(0)$.
- Compute derivative $\sigma'(x)$ in terms of $\sigma(x)$.
- Find $\sigma'(0)$.

Q15. Given:

$$\frac{\partial E}{\partial out} = 0.8$$
$$\frac{\partial out}{\partial net} = 0.3$$
$$\frac{\partial net}{\partial w} = 0.6$$

Compute:

$$\frac{\partial E}{\partial w}$$

Q16. A regression model is:

$$y = 4.162 + 15.509x$$

- Predict y when $x = 2$.
- Predict y when $x = 5$.
- Interpret slope coefficient.

Q17. Prove that for any scalar α and vector x, the 2-norm satisfies:

$$\|\alpha x\|_2 = |\alpha| \|x\|_2$$

Q18. Derive the derivative of Mean Squared Error (MSE) with respect to prediction $\hat{y}$.

Q19. Explain why accuracy can be misleading in highly imbalanced datasets using a numerical example.

Q20. Given:

$$E = \frac{1}{2}(t - o)^2$$

$$o = \sigma(net)$$

$$net = wx + b$$

Derive:

$$\frac{\partial E}{\partial w}$$

Q21. Initial:

$$W_0 = 4$$

Learning rate:

$$\eta = 0.2$$

Gradients over 3 iterations:

$$g_1 = 1.5, \quad g_2 = 0.5, \quad g_3 = -0.2$$

Compute W_3.

Q22. Consider a 5-layer neural network where each hidden layer uses the sigmoid activation function:

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

The derivative of the sigmoid function is given by:

$$\sigma'(x) = \sigma(x)\big(1 - \sigma(x)\big)$$

(a) Using the chain rule, express the gradient of the loss with respect to weights in the first layer in terms of successive derivatives.

(b) Explain mathematically why repeated multiplication of sigmoid derivatives across multiple layers may cause gradients to become very small.

(c) Discuss how this affects learning in deep networks.

Q23. Consider a binary classification problem with the following class distribution:Class 0: 990 samples and Class 1: 10 samples

Suppose a model predicts all samples as Class 0.

(a) Compute the accuracy of the model.

(b) Compute the total number of misclassified samples.

(c) Critically explain why the model is practically useless despite its accuracy.